14

Extrusion

Extrusion is a process which combines several unit operations including mixing, cooking, kneading, shearing, shaping and forming. Extruders are classified according to the method of operation (cold extruders or extruder-cookers) and the method of construction (single- or twin-screw extruders). The principles of operation are similar in all types: raw materials are fed into the extruder barrel and the screw(s) then convey the food along it. Further down the barrel, smaller flights restrict the volume and increase the resistance to movement of the food. As a result, it fills the barrel and the spaces between the screw flights and becomes compressed. As it moves further along the barrel, the screw kneads the material into a semi-solid, plasticised mass. If the food is heated above 100ºC the process is known as *extrusion cooking* (or *hot extrusion*). Here, frictional heat and any additional heating that is used cause the temperature to rise rapidly. The food is then passed to the section of the barrel having the smallest flights, where pressure and shearing is further increased. Finally, it is forced through one or more restricted openings (dies) at the discharge end of the barrel As the food emerges under pressure from the die, it expands to the final shape and cools rapidly as moisture is flashed off as steam. A variety of shapes, including rods, spheres, doughnuts, tubes, strips, squirls or shells can be formed. Typical products include a wide variety of low density, expanded snackfoods and ready-to-eat (RTE) puffed cereals (Table 14.1). Developments using combined super-critical fluid technology (Chapter 6) with extruders to produce a new range of puffed products, pasta and confectionery are described by Rizvi *et al*. (1995). Extruded products may be subsequently processed further by drying (Chapter 15), frying (Chapter 17) or packaging (Chapters 24, 25). Many extruded foods are also suitable for coating or enrobing (Chapter 23). Further details of extrusion technology are given by O'Connor (1987).

Cold extrusion, in which the temperature of the food remains at ambient is used to mix and shape foods such as pasta and meat products. Low pressure extrusion, at temperatures below 100ºC, is used to produce, for example, liquorice, fish pastes, surimi and pet foods (Table 14.1 and Section 14.3).

Extrusion cooking is a high-temperature short-time (HTST) process which reduces microbial contamination and inactivates enzymes. However, the main method of

Table 14.1 Examples of extruded foods

Types of product	Examples
Cereal-based products	Expanded snackfoods RTE and puffed breakfast cereals Soup and beverage bases, instant drinks Weaning foods Pre-gelatinised and modified starches, dextrins Crispbread and croutons Pasta products Pre-cooked composite flours
Sugar-based products	Chewing gum Liquorice Toffee, caramel, peanut brittle Fruit gums
Protein-based products	Texturised vegetable protein (TVP) Semi-moist and expanded petfoods and animal feeds and protein supplements Sausage products, frankfurters, hot dogs Surimi Caseinates Processed cheese

Adapted from Harper (1979), Harper (1987), Heldman and Hartel (1997), Jones (1990) and Best (1994).

preservation of both hot- and cold-extruded foods is by the low water activity of the product (0.1–0.4) (Chapter 1), and for semi-moist products in particular, by the packaging materials that are used.

Extrusion has gained in popularity for the following reasons:

- *Versatility*. A very wide variety of products are possible by changing the ingredients, the operating conditions of the extruder and the shape of the dies. Many extruded foods cannot be easily produced by other methods.
- *Reduced costs*. Extrusion has lower processing costs and higher productivity than other cooking or forming processes. Some traditional processes, including manufacture of cornflakes and frankfurters, are more efficient and cheaper when replaced by extrusion (Section 14.3).
- *High production rates and automated production*. Extruders operate continuously and have high throughputs. For example, production rates of up to $315\,kg\,h^{-1}$ for snackfoods, $1200\,kg\,h^{-1}$ for low-density cereals and $9000\,kg\,h^{-1}$ for dry expanded petfoods are possible (Mans, 1982). Details of automatic control of extruders are described by Olkku *et al.* (1980) and Bailey *et al.* (1995).
- *Product quality*. Extrusion cooking involves high temperatures applied for a short time and the limited heat treatment therefore retains many heat sensitive components.
- *No process effluents*. Extrusion is a low-moisture process that does not produce process effluents. This eliminates water treatment costs and does not create problems of environmental pollution.

Extrusion can be seen as an example of a size enlargement process, in which granular or powdered foods are re-formed into larger pieces. Other examples of size enlargement include forming or moulding (Chapter 5) and agglomeration of powders (Chapter 15). Extruders are also used in the plastics industry to produce packaging materials (Chapter 24).

14.1 Theory

Because extrusion involves simultaneous mixing, kneading and cooking, it causes a large number of complex changes to a food, including hydration of starches and proteins, homogenisation, gelation, shearing, melting of fats, denaturation or re-orientation of proteins, plastification and expansion of the food structure. For many years the empirical knowledge of extruder operators outstripped scientific theory of the sequence and nature of these interactions and their effects. However, computer modelling of fluid flow behaviour and heat transfer inside the extruder barrel has more recently led to a greater understanding of the operation of extruders (Kulshreshtha *et al*. (1995), Tan and Hofer (1995), Elsey *et al*. (1997) and Schoner and Moreira (1997). The two factors that most influence the nature of the extruded product are the rheological properties of the food and the operating conditions of the extruder.

14.1.1 Rheological properties of the food

The properties of the feed material have an important influence on the texture and colour of the product; the most important factors are:

- the type of feed materials
- their moisture content
- the physical state of the materials
- their chemical composition, particularly the amounts and types of starches, proteins, fats and sugars
- the pH of the moistened material.

The composition of the feed material, its moisture content and particle size all influence the viscosity of the product in the extruder. From equations (14.1) and (14.2) below, it can be seen that viscosity is a crucial factor that determines the operating conditions of the extruder and hence the product quality. Different types of feed material produce completely different products when the same operating conditions are used in the same extruder. This is because of differences in the type and amounts of starch, proteins, moisture and other added ingredients (for example oil or emulsifier), which result in different viscosities and hence different flow characteristics. Similarly, addition of acids to adjust the pH of the feed material causes changes to starch gelatinisation and unfolding of protein molecules. This in turn changes the viscosity and hence the structure and strength of the extruded product. Differences in sugar content or pH also produce variations in colour due to different extents of Maillard browning reactions.

During extrusion cooking of starch-based foods, added water causes the starch granules to swell and absorb water to become hydrated. Smaller particles, such as flours or grits, are hydrated and cooked more rapidly than larger particles and this in turn also alters the product quality. The increased moisture content and elevated temperatures cause the starch to gelatinise and a viscous plasticised mass is produced (Mercier, 1980). Gelatinisation of starch usually causes an increase in viscosity, but in extrusion cookers the intense shearing action can also break macro-molecules down to smaller units, resulting in a reduction in viscosity.

As the product leaves the die, it is in a glassy state. It expands rapidly and as it cools, the temperature falls below the glass transition state and strands and matrices form that set the structure and determine the product texture (Blanshard, 1995). Results of detailed research on the changes to starch are described by Guy (1993).

The changes in starch solubility under different conditions of temperature and shear rate are monitored by measuring the *Water Absorption Index* (WAI) and the *Water Solubility Characteristic* (WSC). The WSC decreases as the WAI increases. The WAI of cereal products generally increases with the severity of processing, reaching a maximum at 180–200ºC.

Soy proteins, gluten or caseinate molecules unfold in the hot moist conditions to produce a uniform, viscous plasticised mass. The shearing action prevents re-alignment of the molecules until they emerge from the die. Then the expansion and cooling cause the proteins to polymerise, cross-link and re-orient to form a characteristic fibrous structure and set the final texture of the product. The *nitrogen solubility index* is a measure of the extent of protein denaturation. It decreases during extrusion cooking, and feed materials should therefore have largely undenatured proteins.

14.1.2 Operating characteristics

The most important operating parameters in an extruder are:

- temperature
- pressure
- diameter of the die apertures
- shear rate.

The shear rate is influenced by the internal design of the barrel, its length and the speed and geometry of the screw(s).

Most research to model extruders has been done with single-screw machines (Fig. 14.1 and Section 14.2) because twin-screw extruders are substantially more complex. In operation, the single-screw extruder acts as a type of pump, dragging the food through the barrel and increasing the pressure and temperature before the food is forced through the die. For optimum pumping, the food should stick to the barrel and slip freely from the screw surface (Heldman and Hartel, 1997). However, if food slips on the barrel it does not move through the extruder and is simply mixed. For this reason, the barrel wall is often grooved to minimise slipping.

A simplified model for the operation of an extruder, developed by Harper (1981), assumes that the temperature of the food is constant, fluid flow is Newtonian and laminar

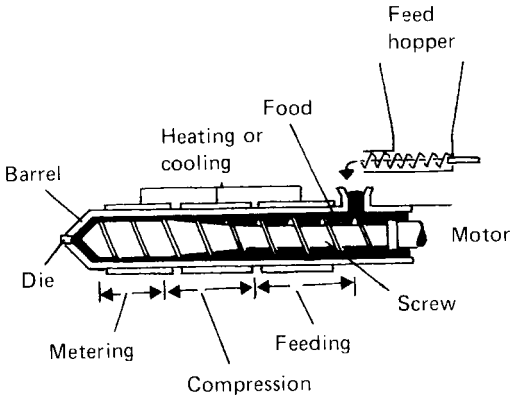


Fig. 14.1 A single-screw extruder.
(Courtesy of Werner and Pfeiderer Ltd.)

(Chapter 1), there is no slip at the barrel wall and no leakage between the screw and the barrel. With these assumptions, the flow through a single screw extruder is calculated using equation (14.1):

$$Q = G_1 N F_d + G_2/\mu . \Delta P/L . F_p \qquad 14.1$$

where $Q\,(m^3\,s^{-1})$ = volumetric flow rate in the metering section (Fig. 14.1), N (rpm) = screw speed, $\mu\,(N\,s\,m^{-2})$ = viscosity of the fluid in the metering section, ΔP (Pa) = pressure increase in the barrel, G_1 and G_2 = constants that depend on screw and barrel geometry, L (m) = length of extruder channel and F_d and F_p = shape factors for flow due to drag and pressure respectively.

The first part of the equation represents fluid flow down the barrel caused by pumping and drag against the barrel wall, whereas the second part represents backward flow from high pressure to low pressure, caused by the increase in pressure in the barrel. Clearly, the amount of pressure in the barrel depends in part on the size of the dies; if the barrel is completely open at the die end, there will be no pressure build-up and the extruder will simply act as a screw conveyor. Conversely, if the die end is completely closed, the pressure will increase until backward flow equals drag flow and no further movement will occur. The extruder would become a mixer. In between these two extremes, the size of the die greatly affects the performance of the extruder. The ratio of pressure to drag flow is known as the *throttling factor (a)* which varies from zero (open die hole) to 1 (die hole is closed). In practice, most extruders operate with a values of between 0.2 and 0.5 (Heldman and Hartel, 1997).

The flow through the die is found using equation (14.2):

$$Q = K' . \Delta P/\mu \qquad 14.2$$

where $Q\,(m^3\,s^{-1})$ = volumetric flow rate through the die (Fig 14.1), $\mu\,(N\,s\,m^{-2})$ = viscosity of the fluid in the die, ΔP (Pa) = pressure drop across the die (from inside the barrel to atmospheric pressure) and K' = a flow resistance factor that depends on the number, shape and size of the die holes, usually found experimentally.

The operating conditions for the extruder can be found by calculating the flow rate and die pressure drop that satisfy both equations. This is shown diagrammatically in Fig. 14.2. The position of the operating point is determined by the type of die, depth of flights on the screw, length and speed of the screw.

It should be noted that the above formulae are based on simple models that do not take account of, for example, leakage of food between the flights and the barrel, a partially empty barrel, changes in temperature and the effects of non-Newtonian fluids. One such fluid is a starch slurry which becomes gelatinised during extrusion and its rheological properties change as it moves through the extruder. It is important that the screw speed (and hence the residence time) is balanced with the extent of heating to obtain the required characteristics in the final product, but this may be very difficult to model as changes in non-Newtonian fluids are significantly more complicated. In practice, therefore, the assumptions made in the formulae may limit their usefulness for predicting flow behaviour or operating conditions, but may be used as a starting point for experimental studies. Most extruder manufacturers use a combination of more sophisticated models and practical experience of the relationships between die shape, extruder construction and characteristics of the product to design their equipment. The situation with twin-screw extruders is even more complex: changes to the degree of inter-meshing of the screws (Section 14.2) or the direction of rotation dramatically alter the flow characteristics in the extruder and make modelling equations very complex.

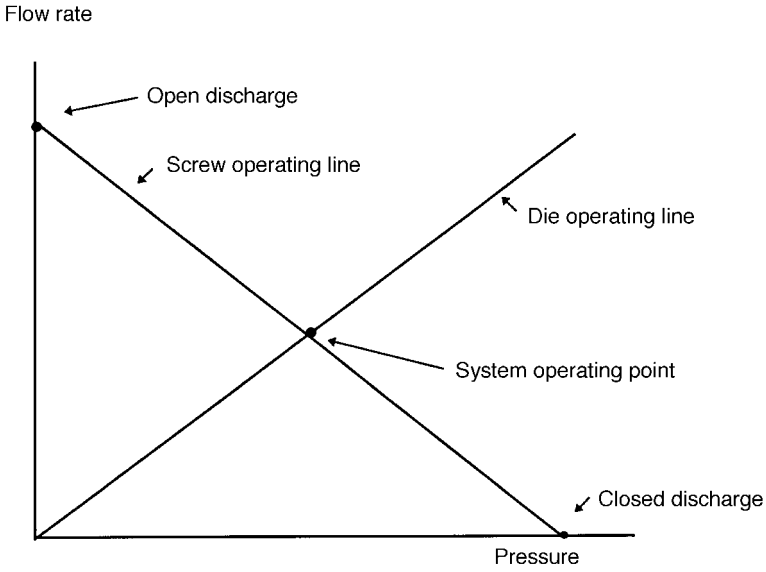


Fig. 14.2 Determination of operating point for a single screw extruder. (Adapted from Miller (1990).)

The rate of heat transfer between a heated barrel jacket and the food during extrusion cooking is found using equation (1.22) in Chapter 1. Related sample problems are found in Chapter 1 (Sample problems 1.7 and 1.8).

14.2 Equipment

14.2.1 Single-screw extruders

The equipment (Fig. 14.1) consists of a cylindrical screw that rotates in a grooved cylindrical barrel, made from hard alloys or hardened stainless steel to withstand the frictional wear. The length to diameter ratio of the barrel is between 2:1 and 25:1 (Hauck, 1993). The pitch and diameter of the screw, the number of flights and the clearance between the flights and the barrel can each be adjusted to change the performance of the extruder. The screw is driven by a variable speed electric motor that is sufficiently powerful to pump the food against the pressure generated in the barrel. The screw speed is one of the main factors that influences the performance of the extruder: it affects the residence time of the product, the amount of frictional heat generated, heat transfer rates and the shearing forces on the product. Typical screw speeds are 150–600 rpm, depending on the application. Compression is achieved in the extruder barrel by back pressure, created by the die and by:

- increasing the diameter of the screw and decreasing the screw pitch
- using a tapered barrel with a constant or decreasing screw pitch
- placing restrictions in the screw flights.

Die pressures vary from around 2000×10^3 Pa for low viscosity products to $17\,000 \times 10^3$ Pa for expanded snackfoods (Heldman and Hartel, 1997).

Single-screw extruders can be classified according to the extent of shearing action on the food into:

Table 14.2 Operating data for different types of extruder

Parameter	High shear	Medium shear	Low shear
Net energy input to product (kWh kg^{-1})	0.10–0.16	0.02–0.08	0.01–0.04
Barrel length:diameter (L/D)	2–15	10–25	5–22
Screw speed (rpm)	> 300	> 200	> 100
Maximum barrel temperature (°C)	110–180	55–145	20–65
Maximum product temperature (°C)	149	79	52
Maximum barrel pressure (kPa)	4000–17 000	2000–4000	550–6000
Product moisture (%)	5–8	15–30	25–75
Product density (kg/m^3)	32–160	160–500	320–800

Adapted from Hauck (1993) and Harper (1979).

- *High shear*. High speeds and shallow flights create high pressures and temperatures that are needed to make breakfast cereals and expanded snackfoods.
- *Medium shear*. For breadings, texturised proteins and semi-moist petfoods.
- *Low shear*. Deep flights and low speeds create low pressures for forming pasta, meat products and gums.

Operating data for different types of extruder are given in Table 14.2.

In extrusion cooking, much of the energy from the extruder motor is lost as friction and rapidly heats the food (between 50 and 100% of total energy input (Harper, 1987)). Additional heating can be achieved using a steam-jacketed barrel and/or by a steam-heated screw (in some applications the jacket is also used to cool the product using cold water). In other designs, electric induction heating elements are used to heat the barrel directly. Some products also require the extruder die to be heated to maintain the viscosity and degree of expansion, whereas others require the die to be cooled to reduce the amount of expansion. The temperature and pressure profiles in different sections of a high-shear cooking extruder are shown in Fig. 14.3.

Single-screw extruders have lower capital and operating costs and require less skill to operate and maintain than twin-screw machines do. They are used for straightforward cooking and forming applications, when the flexibility of a twin-screw machine is not needed.

14.2.2 Twin-screw extruders

The screws in twin-screw extruders rotate within a 'figure of 8' shaped bore in the barrel. Screw length to diameter ratios are between 10:1 and 25:1 (Harper, 1987). Extruders are classified according to the direction of rotation and the way in which the screws intermesh. Co-rotating intermeshing screws, which are self-wiping (the flights of one screw sweep food from the adjacent screw) are most commonly found in food-processing applications (Fig. 14.4). The spacing between the flights can be adjusted so that large spaces initially convey the material to the cooking section and then smaller spaces compress the plasticised mass before extrusion through an inter-changeable die. One of the main advantages of twin-screw extruders is the greater flexibility of operation that is possible by changing the degree of intermeshing of the screws, the number of flights or the angle of pitch of the screw. 'Kneading discs' can also be fitted to the screws so that

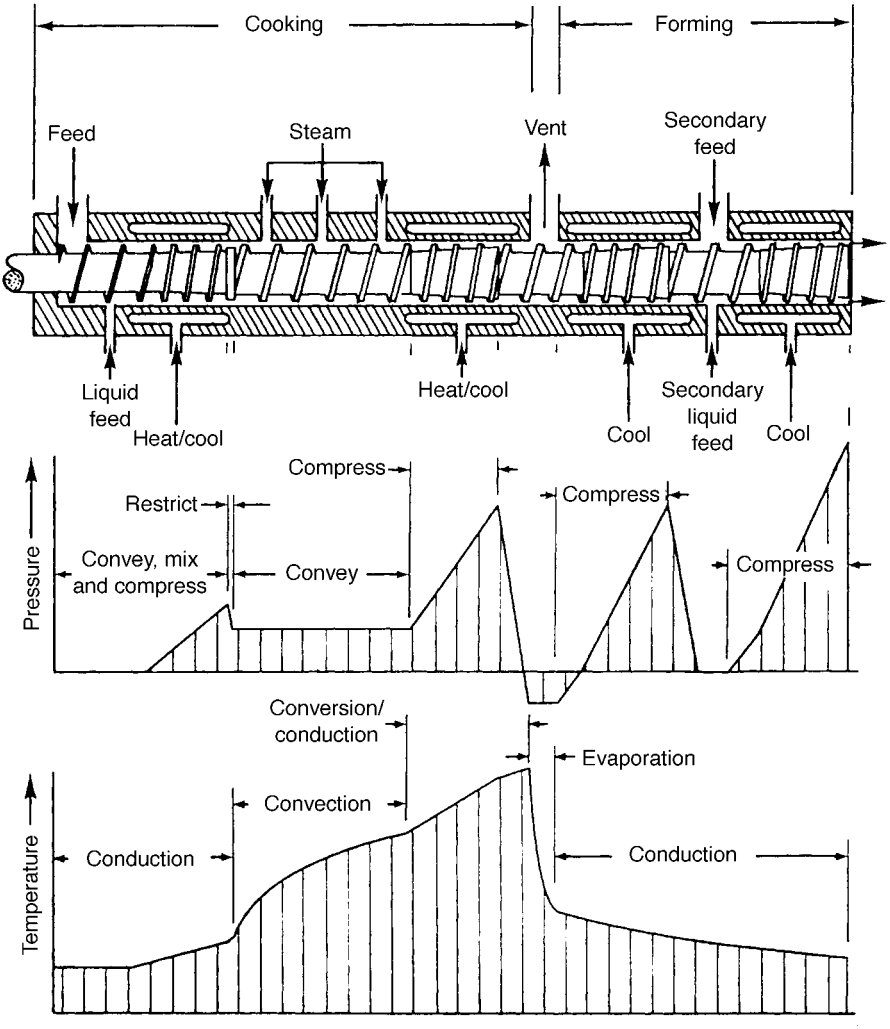
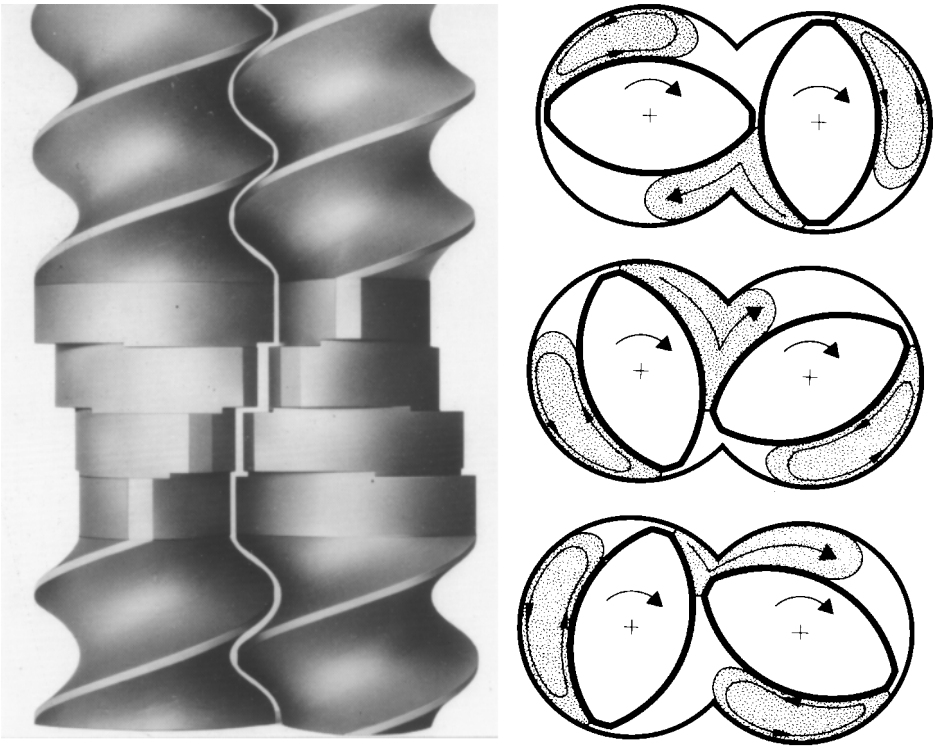


Fig. 14.3 Changes in temperature and pressure in a high shear cooking extruder for expanded food products.
(From Miller (1990).)

the product passes between and through the discs to increase the kneading action. Their operating characteristics are described by Frame (1994).

Twin-screw extruders have the following advantages:

- The throughput is independent of feedrate, and fluctuations in production rate can be accommodated by the positive displacement action of the screws. In contrast, a single screw must be full of material to operate effectively. The positive displacement also produces higher rates of heat transfer and better control of heat transfer than a single screw does.
- Twin-screw machines handle oily, sticky or very wet materials, or other products that slip in a single screw. The limitations for single- and twin-screw machines are respectively 4% and 20% fat, 10% and 40% sugar, and 30% and 65% moisture. There is therefore greater flexibility in operation using different raw materials.



(a) (b)

Fig. 14.4 Kneading elements of a co-rotating twin-screw extruder showing dough mixing: (a) sealing profile; (b) movement of material.
(Courtesy of Werner and Pfeiderer Ltd.)

- Forward or reverse conveying is used to control the pressure in the barrel. For example, in the production of liquorice and fruit gums, the food is heated and compressed by forward conveying, the pressure is released by reverse conveying, to vent excess moisture or to add additional flavour ingredients, and the food is then recompressed for extrusion.
- A short discharge section develops the pressure required for extrusion and thus subjects a smaller part of the machine to wear than in single-screw extruders.
- A mixture of particle sizes, from fine powders to grains, may be used, whereas a single screw is limited to a specific range of granular particle sizes.

The main limitations of twin-screw extruders are the relatively high capital and maintenance costs (up to twice the cost of single-screw equipment) and the greater constraints on operating ranges (Harper, 1987). The complex gearbox that is needed to drive the twin screws results in limitations on the maximum torque, pressure and thrust that can be achieved.

14.2.3 Ancillary equipment

Powders and granular feed materials are first blended with water, steam or other liquids in a preconditioner, to moisten them before feeding into the extruder (Fig. 14.5). This

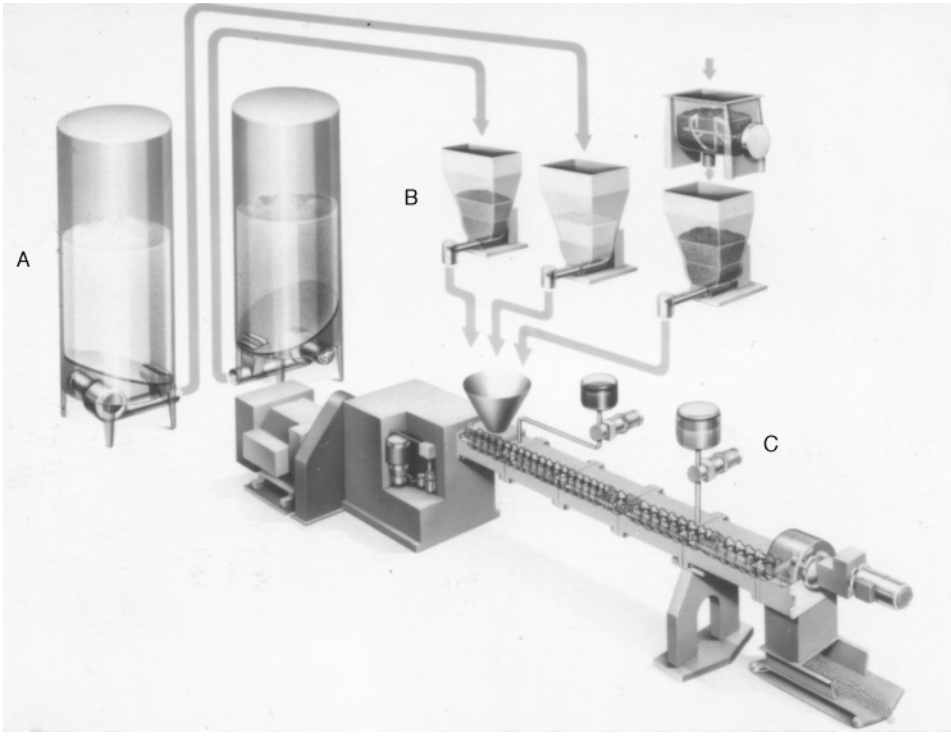


Fig. 14.5 Ancillary equipment with a twin-screw extruder: (A) pre-conditioners; (B) feed hoppers; (C) positive displacement pumps.
(Courtesy of Werner and Pfeiderer Ltd.)

produces a more uniform feed material that can be more accurately metered and provides more uniform extrusion conditions. The residence time in either batch or continuous preconditioners is closely controlled to ensure that each particle is uniformly blended with the liquid, and for steam conditioning, that there is uniform temperature equilibration. Preconditioning with steam or hot water for up to 4.5 mins increases the feed temperature and moisture content, gelatinises starch and denatures proteins. This improves the extruder efficiency, lowers specific energy consumption and reduces equipment wear and maintenance costs (Bailey *et al.*, 1995).

Extruders are fitted with feed hoppers that have screw augers or vibrating feeders to load material at a uniform rate into the barrel. The weight of product on a feeder (or loss in weight of the hopper) is used to automatically control the feedrate. Other control methods include speed control of variable speed motors, the ratio of fluid feedrate to dry material feedrate and barrel temperature. Bailey *et al.* (1995) describe a computerised system of process control that is able to control the extrusion process, the start-up and shut-down sequences, alarm recognition, storage of formulations and presentation of colour displays of the process to the operator. The control system continuously monitors over one hundred process alarm conditions, including the product formulation in relation to the operating conditions. It visually alerts the operator if non-specified conditions exist, and in extreme cases, controls an orderly shutdown or an emergency stop. Another system for control of a twin-screw extruder is descibed by Lu *et al.* (1993) (also Chapter 2).

Liquids and slurries are pumped into the extruder barrel by positive displacement pumps, with flow measuring devices fitted in-line to control the feedrate. Dies are produced with different shaped holes and are readily inter-changeable. Shapes may be simple, such as round holes to produce rods, square holes for bars or slots to produce sheets; or they may have more complex patterns for specially shaped products. Extruders may also be fitted with a special die to continuously inject a filling into an outer shell. This is known as *co-extrusion* and is used, for example, to produce filled confectionery.

After material leaves the die, it is cut into the required lengths by a series of knives that rotate across the face of the die. The speed of rotation is adjusted to the throughput to produce the correct length. Alternatively, the product may be transported by conveyor to a separate guillotine for cutting.

14.3 Applications

14.3.1 Cold extrusion

In this process, the product is extruded without cooking or distortion of the food. The extruder has a deep-flighted screw, which operates at a low speed in a smooth barrel, to knead and extrude the material with little friction. It is used to produce pasta, hot dogs, pastry doughs and some types of confectionery. Typical operating conditions are shown in Table 14.2 for low shear conditions.

14.3.2 Extrusion cooking

High pressures and temperatures are used to form expanded products. The rapid release of pressure as the food emerges from the die causes instantaneous expansion of steam and gas in the material, to form a low-density product. Hot extrusion is a HTST process, which minimises the loss of nutrients and reduces microbial contamination. The moisture content of some products, for example snackfoods, crispbread and breakfast cereals, is further reduced after extrusion by drying (Chapters 14 and 18).

Confectionery products

HTST extrusion cooking is used to produce a gelatinised, chewy product such as fruit gums and liquorice, from a mixture of sugar, glucose and starch. The heat gelatinises the starch, dissolves the sugar and vaporises excess water which is vented from the machine. Colourings and flavours are added to the plasticised material and, after mixing, it is cooled and extruded. The product texture can be adjusted from soft to elastic by control over the formulation and processing conditions, the shape can be changed by changing the die, and a variety of flavours and colours may be added. These different combinations permit a very large range of potential products, including liquorice, toffee, fudge, boiled sweets, creams, and chocolate, each produced by the same equipment (Best, 1994). Product uniformity is high, no after-drying is required, and there is a rapid start-up and shut-down.

Hard-boiled sweets are produced from granulated sugar and corn syrup. The temperature in the extruder is raised to 165ºC to produce a homogeneous, decrystallised mass. Acids, flavours and colour are added to the sugar mass, and the moisture content is reduced to 2% as the product emerges from the die into a vacuum chamber. It is then fed to stamping or forming machines to produce the required shape. Compared with traditional methods which use boiling pans (Chapter 13), energy consumption in an

extruder operating at 1000 kg h^{-1} is reduced from 971 to 551 kJ per kilogram of sugar mass, and steam consumption is reduced from 0.485 to 0.193 kg per kilogram of sugar mass (Huber, 1984).

Cereal products

Crispbread

Wheat flour, milk powder, corn starch, sugar and water are mixed and the product is extruded at a high temperature and pressure. The crispbread is then toasted to reduce the moisture content further and to brown the surface. Savings compared with oven baking are up to 66% in energy consumption, as less moisture is removed, and up to 60% in capital costs and floor space, as large ovens are unnecessary (Vincent, 1984).

A lower pressure and/or larger die aperture are used to produce *prefoms* or *half-products* from pre-gelatinised cereal doughs. These small, hard, dense pellets are suitable for extended storage and transport to other processors, where the final product is produced by frying, toasting or puffing. When the half-products are heated in air or oil, they are softened and develop the necessary physical properties for expansion. The residual moisture in the pellets then turns to steam, to expand the product rapidly to its final shape.

Breakfast cereals

In traditional cornflake manufacture, large maize kernels (grits) were needed, as the size of the individual grit determined the size of the final cornflake. Grits were then pressure cooked, dried, tempered to ensure a uniform moisture distribution, flaked, toasted and sprayed with a vitamin solution. The total processing time exceeded 5 h. Dough pellets are now produced in a low-pressure extruder. The size of the pellets determines the size of the cornflakes. They are then flaked, toasted and sprayed as before. The advantages of extrusion cooking are:

- a reduction in raw material costs (19.4%) as maize grits of any size may be used, a reduction in energy consumption (100%), capital expenditure (44%) and labour costs (14.8%) (Darrington, 1987)
- rapid processing to produce cornflakes within minutes of start-up
- close control over the size and quality of the final product
- flexibility to change the product specification easily (Slater, 1984).

Details of the manufacture of ready-to-eat breakfast cereals are given by Bailey *et al.* (1995) and are shown in Fig. 14.6. Each stage of feed material preparation, extrusion, drying, separation of flakes that are broken or stuck together (in the 'scalper') and coating is linked by a programmable logic controller (Chapter 2).

Protein-based foods

Texturised vegetable protein (TVP)

Extrusion cooking destroys the enzymes present in soybeans, including a urease which reduces the shelf life, a lipoxidase which causes off flavours by oxidation of soya oil and also a trypsin inhibitor which reduces protein digestibility. This improves the acceptability, digestibility and shelf life of the product. Defatted soya flour, soya concentrate or isolate are moistened and the pH is adjusted. A lower pH (5.5) increases chewiness in the final product, whereas a higher pH (8.5) produces a tender product and more rapid rehydration. Colours, flavours and calcium chloride firming agent are added, and the material is plasticised in an extruder at 60–104°C. It is then extruded to form

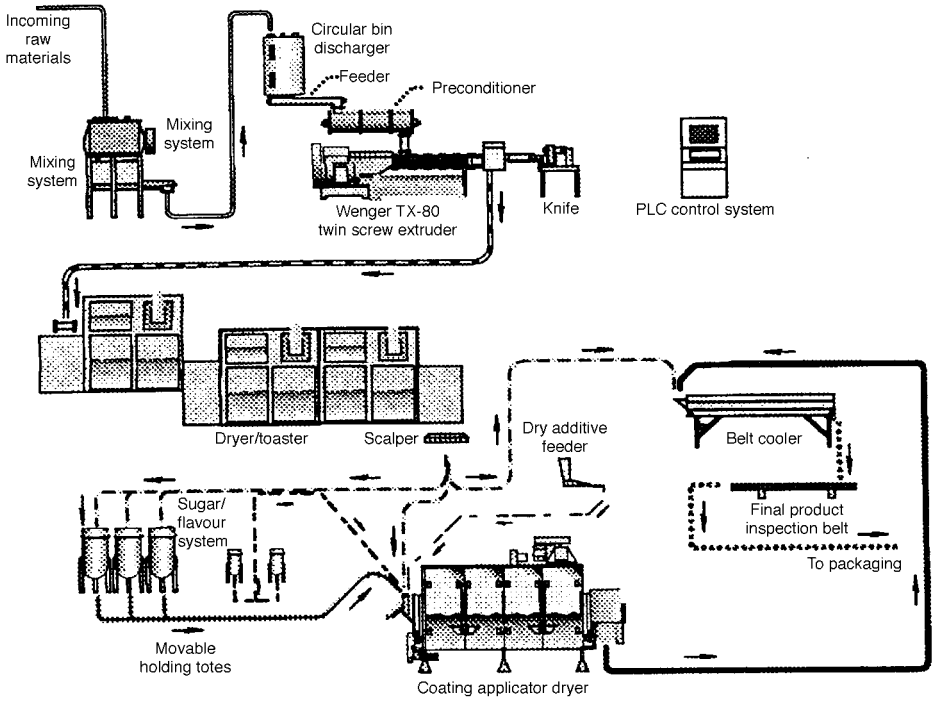


Fig. 14.6 Production of ready-to-eat breakfast cereal.
(From Bailey *et al.* (1995).)

expanded texturised strands, which are cooled and dried to 6–8% moisture content. Details of the production of different texturised soya products are given by Smith and Ben-Gera (1980).

Meat and fish products

Application of extruders to meat and fish products has mostly focused on production of extruded snacks or shelf-stable starch pellets that incorporate previously under-utilised by-products from meat, fish or prawns. Additionally, the manufacture of shiozuri surimi from ground, minced fish has been investigated using extruders operating at a die temperature of 6–27ºC. A detailed description of surimi processing and the application of the process to red meats and poultry is given by Knight *et al.* (1991).

Other developments

A number of more recent applications of extrusion are described by Jones (1990), including the use of extruders as enzymatic reactors with thermostable α-amylase to produce modified starches (Chapter 7). Caseinates are also subjected to partial hydrolysis in an extruder with selected proteases and the products are reported to have very good bacteriological quality, improved colour, flavour and water absorption properties. Another application is the direct production of oligosaccharide mixtures from potato starch, without the use of enzymes, that are suitable for dietetic and infant food. Oil can be extracted more efficiently from soybean flour by first using dry extrusion (10–14% moisture) at 135ºC followed immediately by pressing in a continuous screw press (Chapter 6). The process is said to produce very high quality oil, similar to a refined,

deodorised product, together with an oilcake, containing 50% protein and 90% inactivation of trypsin inhibitors, that is suitable for human consumption.

Extruders are also beginning to be used for decontamination of spices and for sterilisation of cocoa nibs prior to roasting for chocolate manufacture. In the latter application, it is reported that extrusion cooking results in a 1000-fold reduction in micro-organisms and removal of off-flavours that eliminates the need for a time consuming and expensive conching stage (also Chapter 23).

14.4 Effect on foods

14.4.1 Sensory characteristics

Production of characteristic textures is one of the main features of extrusion technology. The extent of changes to starch, determined by the operating conditions and feed materials (Section 14.1), produce the wide range of product textures that can be achieved. The HTST conditions in extrusion cooking have only minor effects on the natural colour and flavour of foods. However, in many foods the colour of the product is determined by the synthetic pigments (Appendix C) added to the feed material as water- or oil-soluble powders, emulsions or lakes. Fading of colour due to product expansion, excessive heat or reactions with proteins, reducing sugars or metal ions may be a problem in some extruded foods. Added flavours are mixed with ingredients before cold extrusion, but this is largely unsuccessful in extrusion cooking as the flavours are volatilised when the food emerges from the die. Micro-encapsulated flavours are more suitable but expensive. Flavours are therefore more often applied to the surface of extruded foods in the form of sprayed emulsions or viscous slurries. However, this may cause stickiness in some products and hence require additional drying.

14.4.2 Nutritional value

Vitamin losses in extruded foods vary according to the type of food, the moisture content, the temperature of processing and the holding time. Generally, losses are minimal in cold extrusion. The HTST conditions in extrusion cooking, and the rapid cooling as the product emerges from the die, cause relatively small losses of most vitamins and essential amino acids. For example at an extruder temperature of 154ºC there is a 95% retention of thiamin and little loss of riboflavin, pyridoxine, niacin or folic acid in cereals. However, losses of ascorbic acid and β-carotene are up to 50%, depending on the time that the food is held at the elevated temperatures (Harper, 1979), and loss of lysine, cystine and methionine in rice products varies between 50 and 90% depending on processing conditions (Seiler, 1984). In soy flour the changes to proteins depend on the formulation and processing conditions. High temperatures and the presence of sugars cause Maillard browning and a reduction in protein quality. Lower temperatures and low concentrations of sugars result in an increase in protein digestibility, owing to rearrangement of the protein structure. Destruction of anti-nutritional components in soya products improves the nutritive value of texturised vegetable proteins.

14.5 Acknowledgements

Grateful acknowledgement is made for information supplied by: Werner and Pfeiderer

(UK) Ltd, Stockport SK6 6AG, UK; Vincent Processes Ltd, Shaw, Newbury, Berkshire RG13 2NT, UK; Baker Perkins Ltd, Peterborough PE3 6TA, UK.

14.6 References

BAILEY, L. N., HAUCK, B. W., SEVATSON, E. S. and SINGER, R. E. (1995) Ready-to-eat breakfast cereal production. In: A. Turner (ed.) *Food Technology International Europe*, Sterling Publications International, London, pp. 127–132.

BEST, E. T. (1994) Confectionery extrusion. In: N. D. Frame (ed.) *The Technology of Extrusion Cooking*. Blackie Academic and Professional, Glasgow, pp. 190–236.

BLANSHARD, J. M. V. (1995) The glass transition. In: S. Beckett (ed.) *Physico-Chemical Aspects of Food Processing*. Blackie, London.

DARRINGTON, H. (1987) A long-running cereal. *Food Manuf.* **3**, 47–48.

ELSEY, A. U., RIEPENHAUSEN, J., McKAY, B., BARTON, G. W. and WILLIS, M. (1997) Modelling and control of a food extrusion process. *Computers and Chemical Engineering*, **21** No. SS, pp. S361–366.

FRAME, N. D. (1994) Operational characteristics of the co-rotating twin-screw extruder. In: N. D. Frame (ed.) *The Technology of Extrusion Cooking.* Blackie Academic and Professional, Glasgow, pp. 1–51.

GUY, R. (1993) Creating texture and flavour in extruded products. In: A. Turner (ed.) *Food Technology International Europe*. Sterling Publications International, London, pp. 57–60.

HARPER, J. M. (1979) Food extrusion. *CRC Crit. Rev. Food. Sci. Nur.* February 155–215.

HARPER, J. M. (1981) *Extrusion of Foods*, Vol II. CRC Press, Boca Raton, FL.

HARPER, J. M. (1987) High-temperature short-time extrusion cooking. In: A. Turner (ed.) *Food Technology International Europe*. Sterling Publications International, London, pp. 51–55.

HAUCK, B. W. (1993) Choosing an extruder. In: A. Turner (ed.) *Food Technology International Europe*. Sterling Publications International, London, pp. 81–82.

HELDMAN, D. R. and HARTEL R. W. (1997) *Principles of Food Processing*. Chapman and Hall, New York, pp. 253–283.

HUBER, G. R. (1984) New extrusion technology for confectionery products. *Manuf. Confect.* May, 51–52, 54.

JONES, S. A. (1990) New developments in extrusion cooking. In: A. Turner (ed.) *Food Technology International Europe*, Sterling Publications International, London, pp. 54–58.

KNIGHT, M. K., CHOO, B. K. and WOOD, J. M. (1991) Applying the surimi process to red meats and poultry. In: A. Turner (ed.) *Food Technology International Europe*. Sterling Publications International, London, pp. 147–149.

KULSHRESHTHA, M. K., ZAROR, C. A. and JUKES, D. J. (1995) Simulating the performance of a control system for food extruders using model-based set-point adjustment. *Food Control* **6** (3), 135–141.

LU, Q., MULVANEY, S. J., HSEITH, F. and HUFF, H. E. (1993) Model and strategies for computer control of a twin-screw extruder. *Food Control* **4** (1), 25–33.

MANS, J. (1982) Extruders, *Prep. Foods* **11**, 60–63.

MERCIER, C. (1980) Structure and digestibility alterations of cereal starches by twin-screw extrusion-cooking. In: P. Linko, Y. Malkki, J. Olkku and J. Larinkari (eds) *Food Process Engineering*, Vol. 1. Applied Science, London, pp. 795–807.

MILLER, R. C. (1990) Unit operations and equipment IV: extrusion and extruders. In: R. B. Fast and E. F. Caldwell (eds) *Breakfast Cereals and How They are Made*. American Association of Cereal Chemists, Inc., St Paul, MN, pp. 135–196.

O'CONNOR, C. (1987) *Extrusion Technology for the Food Industry*. Elsevier Applied Science, London.

OLKKU, J., HASSINEN, H., ANTILA, J., POHJANPALO, H. and LINKO, P. (1980) Automation of HTST-extrusion cooker. In: P. Linko, Y. Malkki, J. Olkku and J. Larinkari (eds) *Food Process Engineering*, Vol. 1. Applied Science, London, pp. 777–790.

RIZVI, S. S. H, MULVANEY, S. J. and SOKHEY, A. S. (1995) The combined application of supercritical fluid and extrusion technology. *Trends in Food Science & Technology* **6**, July, 232–240.

SCHONER, S. L and MOREIRA, R. G. (1997) Dynamic analysis of on-line product quality attributes of a food extruder. *Food Science & Technol. Int.* **3** (6), 413–421.

SEILER, K. (1984) Extrusion cooking and food processing. *Food Trade Rev.* March 124–125, 127.

SLATER, G. (1984) Application of extrusion to the production of breakfast cereals. *Food. Trade Rev.* March 127–128, 131–132.

SMITH, O. B. and BEN-GERA, I. (1980) The application of high temperature short time extrusion cooking in the food industry. In: P. Linko, Y. Malkki, J. Olkku and J. Larinkari (eds) *Food Process Engineering*, Vol. 1. Applied Science, London, pp. 726–744.

TAN, J. L. and HOFER, J. M. (1995) Self-tuning predictive control of processing temperature for food extrusion. *J. Process Control* **5** (3), 183–189.

VINCENT, M. W. (1984) *Extruded confectionery – equipment and process*. Vincent Processes Ltd, Shaw, Newbury, Berkshire RG13 2NT, UK.